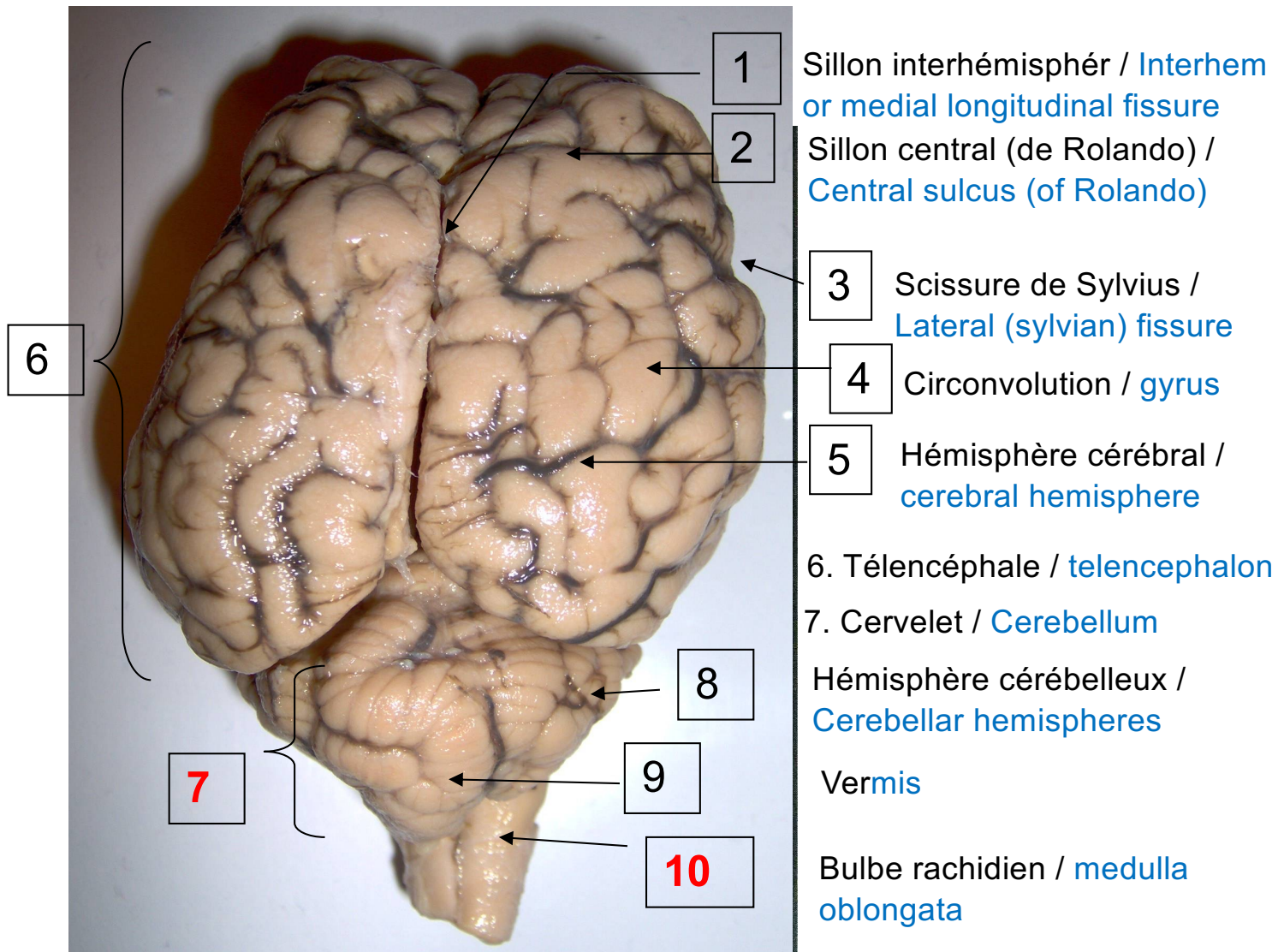


NEURO-ANATOMY

- 1) Give a title and add legends on the following pictures of sheep brains
- 2) What is (are) the function(s) of the structures highlighted in red
- 3) List (in the order of information flow) the nervous system structures involved in the transfer of the following informations: somatosensory, motor, visual and auditory.

Dorsal view

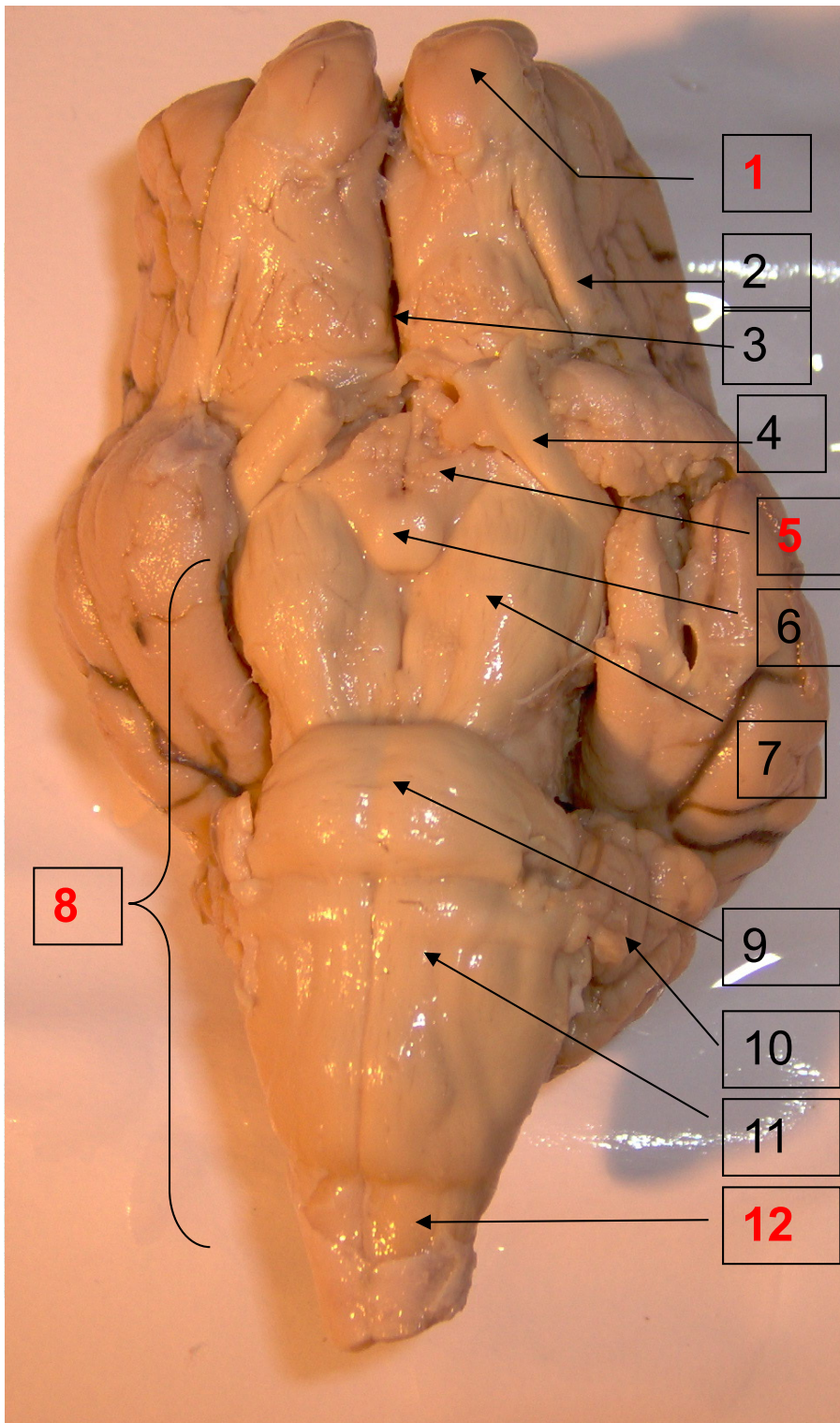


Functions:

7:

10:

Ventral view



1 Bulbe olfactif / Olfact. bulb

2 Tractus olfactif / Olfact. tract

3 Sillon interhémisph./ Interhem. or medial longitudinal fissure

4 Tractus optique / Optic tract

5 Hypothalamus

6 Corps mammillaires / Mammillary bodies

7 Pédonc. Cérébraux / Cerebral peduncles

8. Tronc cérébral / brainstem

9 Pont de Varole / Pons

10 Cervelet / Cerebellum

11 Bulbe rachidien / medulla oblongata

12 Moelle épinière / Spinal cord

Functions:

1:

5:

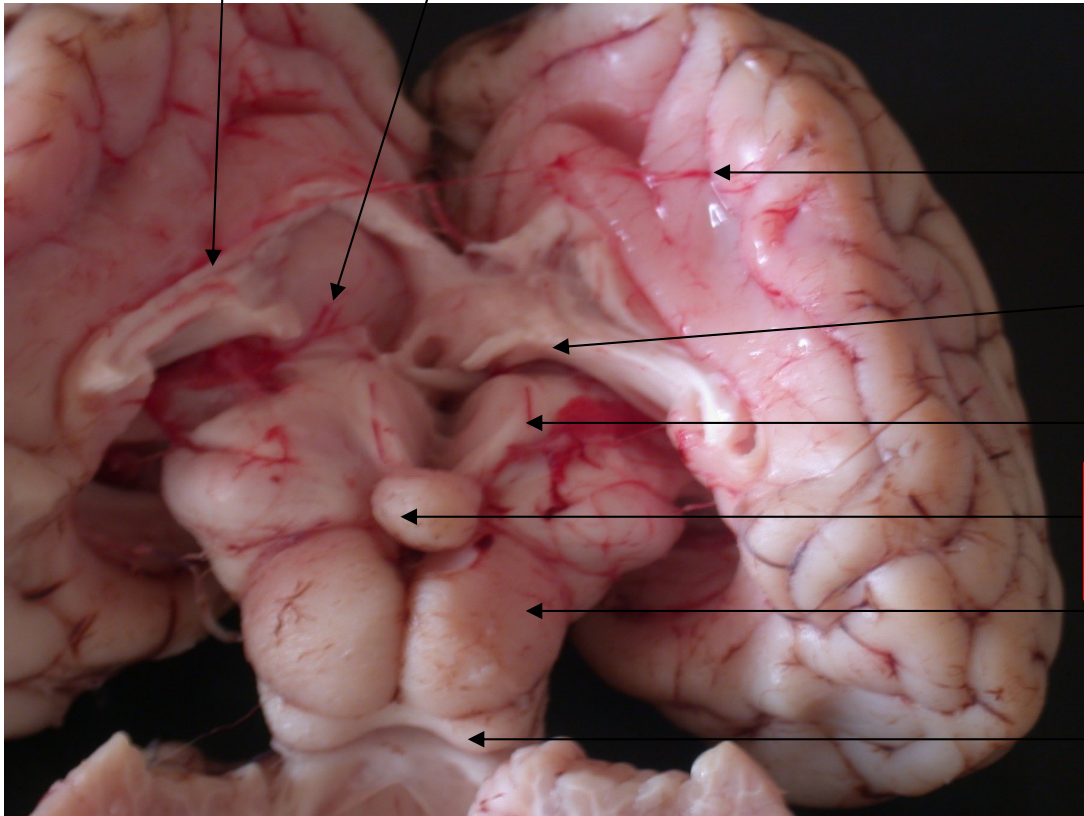
8:

12:

Dorsal view after separation of the two telencephalic hemispheres

1. Corps calleux /
Corpus callosum

2. Striatum



H. cérébral /
cerebral Hem.

Fornix

3. Thalamus

4. Epiphyse /
Pineal gland

TQJ ant. /
Anterior colliculi

TQJ post. /
Posterior colliculi

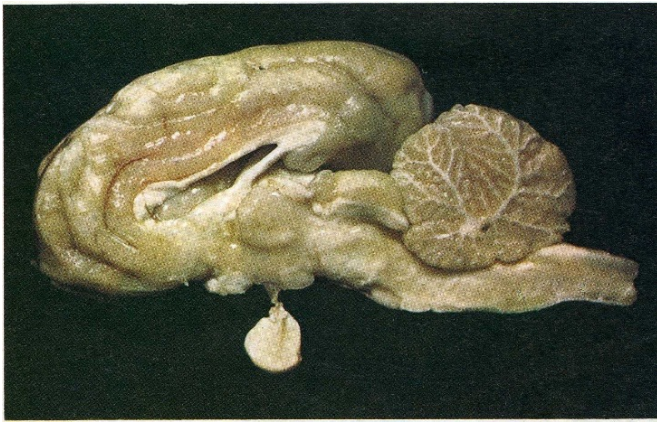
Functions:

1:

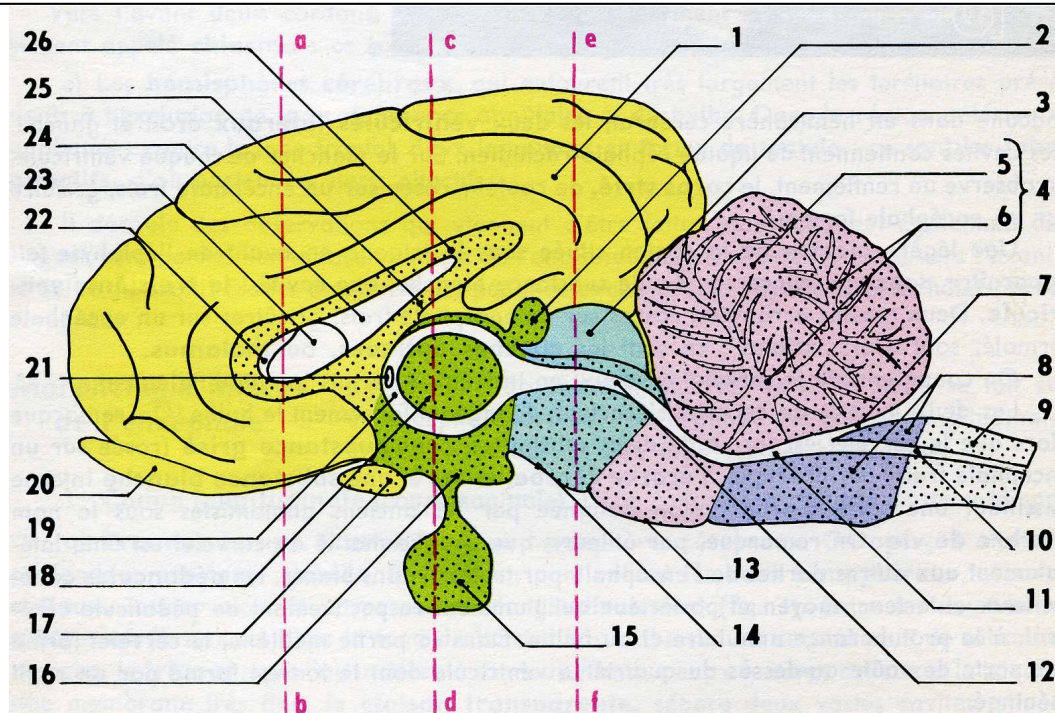
2:

3:

4:



Médio-sagittal section



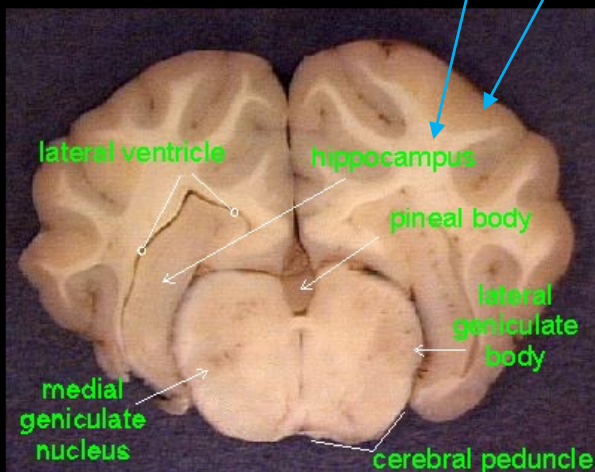
- | | | |
|--|--|---|
| 1 H. cérébral / cerebral hem. | 9 Moelle ép. / Spinal cord | 18 Thalamus |
| 2 Épiphyse / pineal gland | 10 Canal de l'épend. / ependymal (or central) canal | 19 3°V |
| 3 Colliculus | 11 4°V | 20 Bulbe olfactif / Olf. Bulb |
| 4 Vermis | 12 Aqueduc de S. / Lateral (sylvian) fissure | 21 Trou de Monro (Foramen interventriculaire) / Interventr. foramen |
| 5 subst. grise / grey matter | 13 Pont de Varole / Pons | 22 Septum pellucidum |
| 6 subst. Blanche / white matter | 14 Pédoncule cérébral / Cerebral peduncle | 23 Striatum |
| 7 plexus chor. 4° V | 15 Hypophyse / Pineal gland | 24 V. latéral / Lateral Ventr. |
| 8 Bulbe rach. / med. oblongata | 16 hypothalamus | 25 Fornix |
| | 17 Chiasma opt. / Optic Chiasm | 26 Corps calleux / Corpus callosum |

Functions:

7:

White matter

Cerebral cortex



-29-

Plane of Section [Show/Hide](#)

Dissection Instructions: Right Through the Pineal Body--

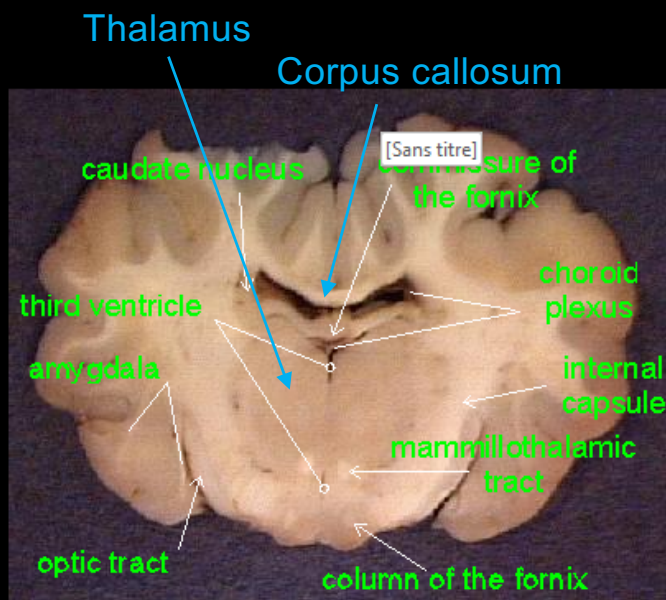
Here we have captured the pineal gland sitting in the notch between the two lobes of the thalamus. The thin 3rd ventricle can be seen (we have passed caudal to the massa intermedia). The medial geniculate nucleus may be visible.

<<< >>>

Options +/- Labels

Behavioral Neuroscience Lab University of Scranton

[Discuss the Guide on Twitter](#)



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Plane of Section

Dissection Instructions: Rostral Mammillary Body--

Not much has changed from the last section.
We have indicated some tufts of choroid plexus in the third and lateral ventricles.

At this point, you should have fibers crossing between the

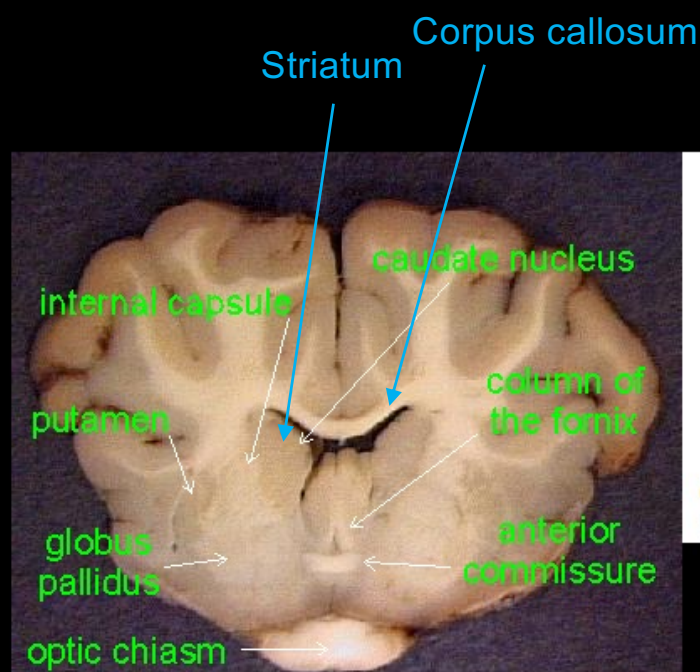
<<< >>>

Options +/- Labels

Coupe frontale passant par les noyaux gris centraux

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-22-

Plane of Section

Dissection Instructions: Anterior Optic Chiasm--

In this section you can see the anterior part of the optic chiasm, which is stuck to the ventral surface of the brain. You can also see the decussating fibers of

<<< >>>

Options

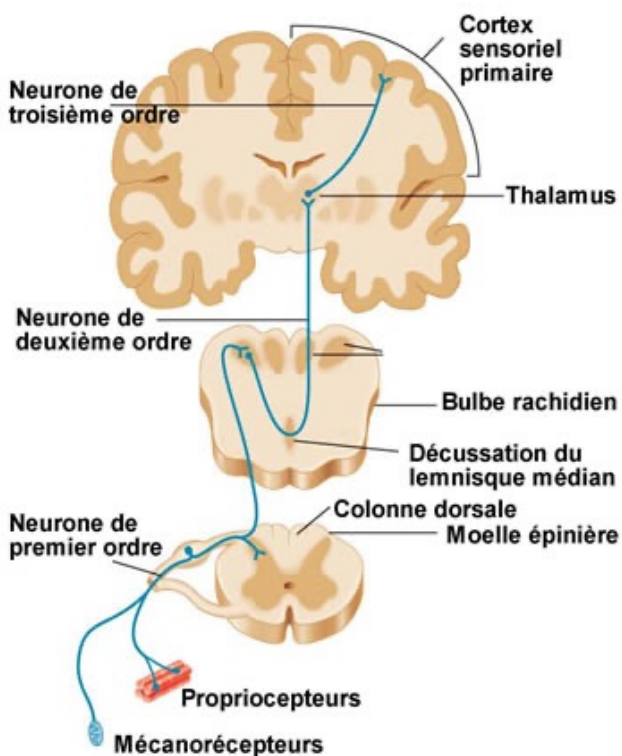
+/- Labels

3) Structures involved in the transfer of the following informations:

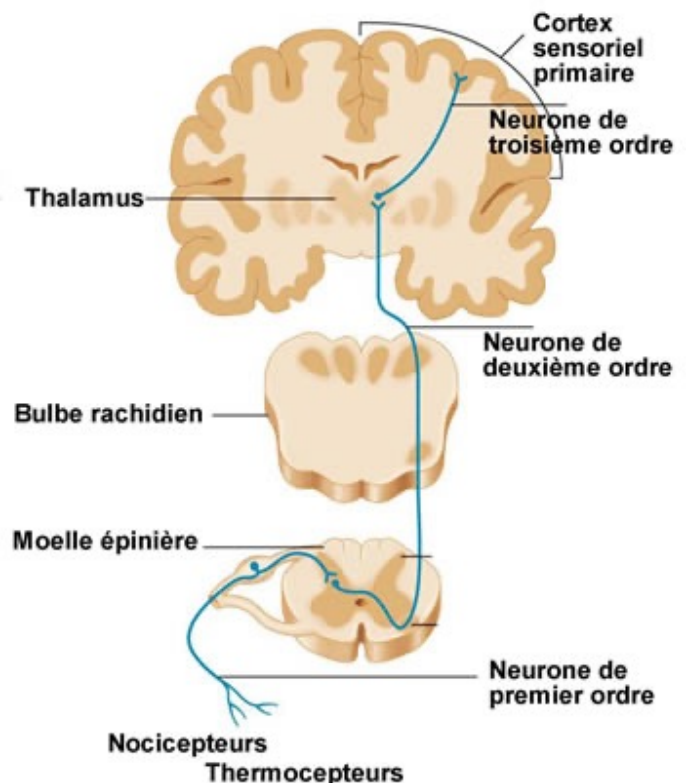
Somatosensory:

PNS: sensory neurons → CNS: spinal cord (spinothalamic tract) ou (via spinal cord) medulla oblongata (lemniscal tract):
2nd order neurons: decussation → Thalamus: 3rd order neurons → primary (somato-) sensory cortex

lemniscal tract

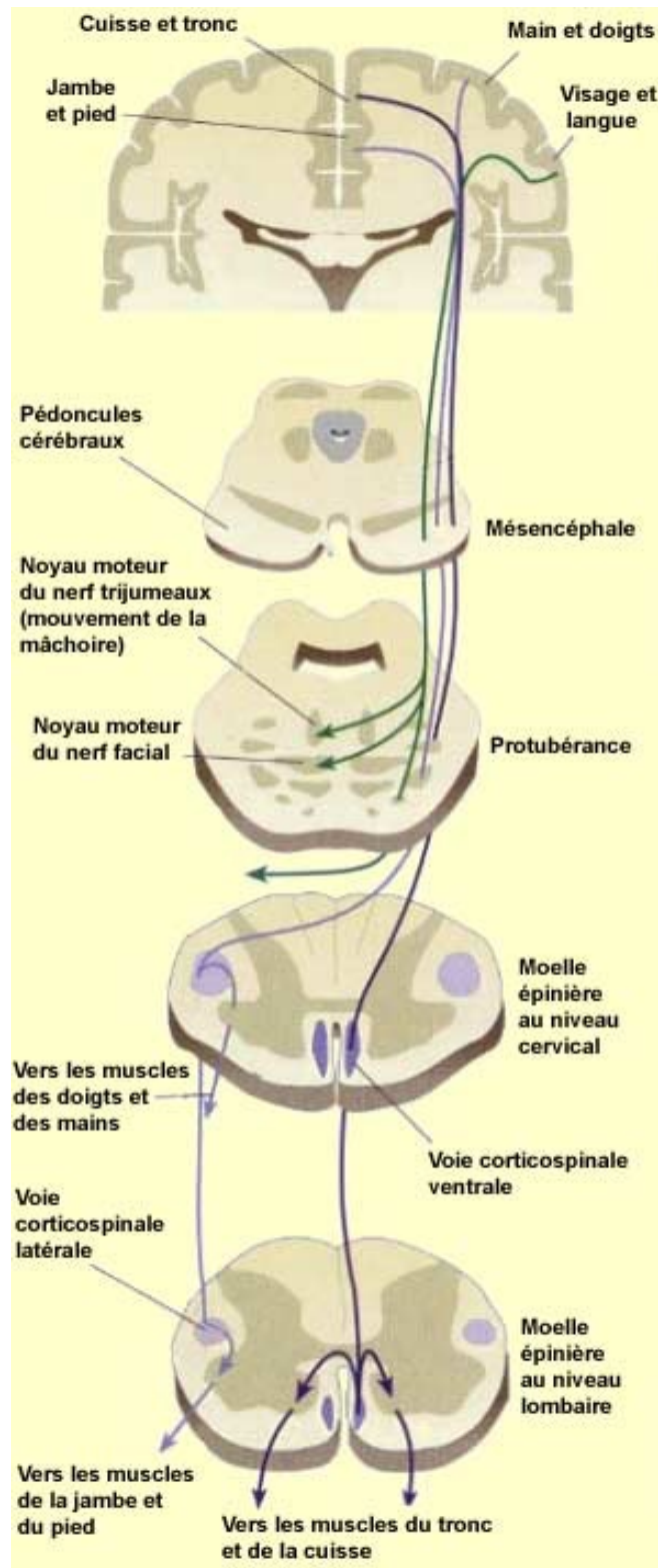


spinothalamic tract



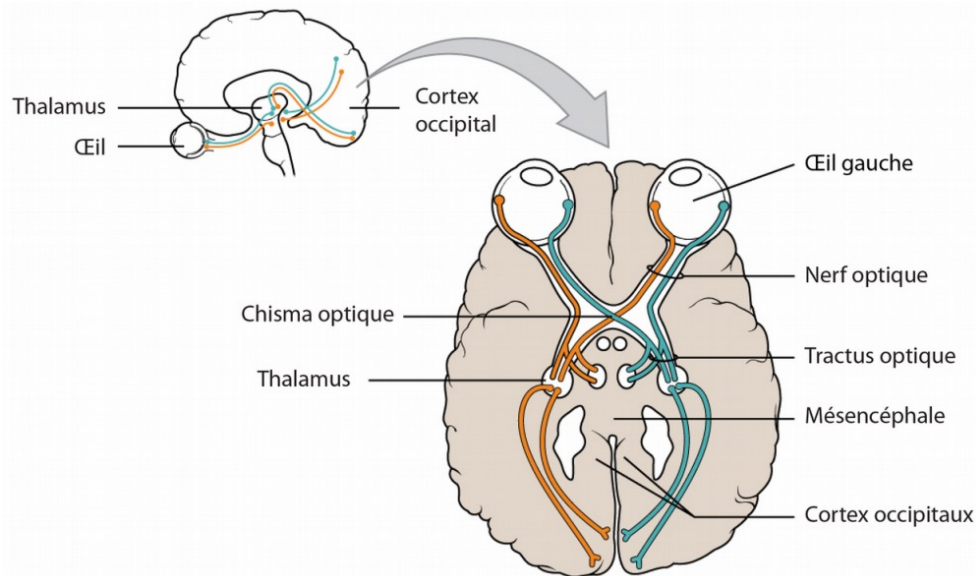
Motor:

Moto cortex → Motor nuclei in brainstem (cortico-bulbar tract) ou spinal cord (cortico-spinal tract) (with decussation at different levels)



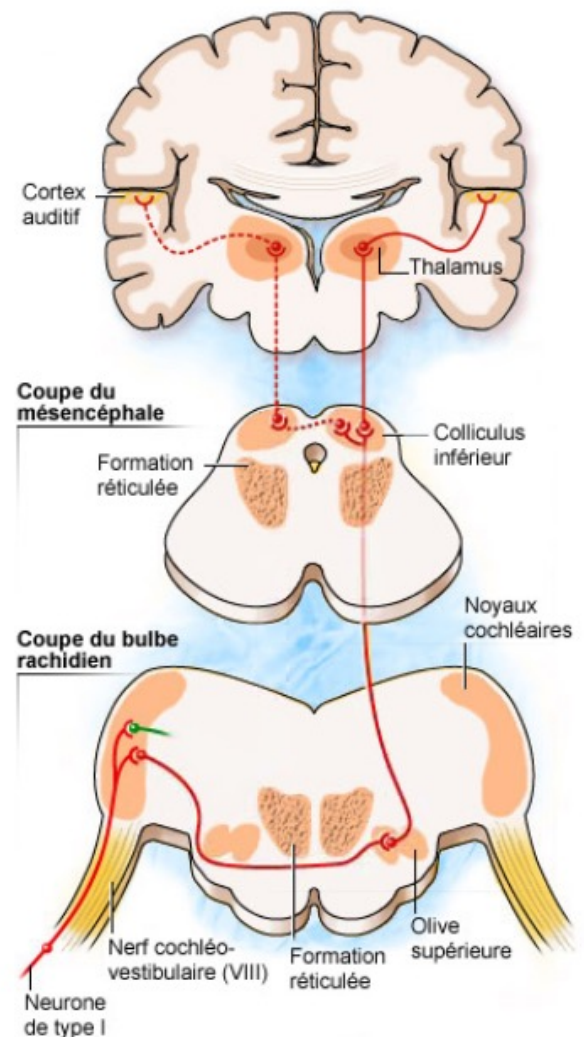
Visual:

Retina (retinal ganglion cells) (via optic nerve and optic chiasm)
→ Thalamus (Lateral geniculate nucleus) → visual cortex



Auditory:

Cochlea (via cochleo-vestibular nerve, VIII) → medulla oblongata (cochlear nuclei) → 1. Mesencephalon (inferior colliculus) and 2. Thalamus (Median geniculate nucleus) → auditory cortex



Preparation of white matter and gray matter smears

Materials

- 1 fresh pig brain per lab group
- 2 glass slides per pair of students
- 2 glass coverslips per pair
- Methylene blue or equivalent
- One plastic pipette per pair
- 1 microscope per pair
- 1 scalpel or knife per pair

Procedure

- Step 1: Using the tip of a scalpel or knife, collect a small amount of white matter (from the corpus callosum)
- Step 2: Place the sample on the slide
- Step 3: Spread with the cover slip
- Step 4: Place a drop of methylene blue on the spread sample.
- Step 5: Place the cover slip on the sample, avoiding air bubbles.
- Step 6: Repeat steps 1 to 5 with a sample from the gray matter (cerebral or cerebellar cortex)

Observation under the microscope

Conclusion for white matter:

Conclusion for gray matter:

Observation of brain sections stained with cresyl-violet

Materials:

- Coronal sections of rat brain stained with cresyl violet*
- Optical microscope
- Stereotaxic atlas

*Histological staining with cresyl violet

The basic dye cresyl violet binds to acidic and basophilic cellular structures, particularly nucleic acids, staining the nucleus and ribosomes in blue-violet.

Cresyl violet staining is a variant of Nissl staining, used specifically for the study of the central nervous system (CNS). This staining highlights the cytoarchitecture of different regions of the central nervous system (gray vs. white matter).

In the gray matter, this staining highlights neurons in relation to glial cells, revealing the ribosomal RNA of the rough endoplasmic reticulum, whose stacking forms the Nissl bodies characteristic of neurons. Thus, neuronal cell bodies appear intensely stained (purple/blue), especially in their cytoplasm, while the nucleus and nucleolus are clearly visible and the axons and most dendrites (which are poor in ribosomes) remain clear.

Procedure:

1. Observe brain sections under the microscope and distinguish gray from white matter, ventricles (lateral or third, if present), based on the staining pattern.
2. Using the stereotaxic atlas, identify the antero-posterior position of the observed sections, based on the morphology of major anatomical structures, such as:
 - Lateral ventricles (disappearing caudally)
 - Corpus callosum (pale, myelinated, supraventricular band)
 - Striatum (lateral, anterior gray matter area)
 - Hippocampus (appears more caudally, "C" shaped structure)
 - Thalamus (median, caudal to the striatum)
3. Observe and compare the cytoarchitecture (neuronal density, somatic size, lamination) of the identified gray matter regions (e.g., hippocampus/cortex).